Screenshots:

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ make
gcc -O2 -Wall -Wextra -o engine engine.c -lpthread
engine.c: In function ‘child_fn’:
engine.c:540:5: warning: ignoring return value of ‘chdir’ declared with attribute ‘warn_unused_result
’ [-Wunused-result]
  540 |     chdir("/");
      |     ^~~~~~~~~~
At top level:
engine.c:282:13: warning: ‘bounded_buffer_begin_shutdown’ defined but not used [-Wunused-function]
  282 | static void bounded_buffer_begin_shutdown(bounded_buffer_t *buffer)
      |             ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~
engine.c:275:13: warning: ‘bounded_buffer_destroy’ defined but not used [-Wunused-function]
  275 | static void bounded_buffer_destroy(bounded_buffer_t *buffer)
      |             ^~~~~~~~~~~~~~~~~~~~~~
gcc -O2 -Wall -static -o memory_hog memory_hog.c
gcc -O2 -Wall -static -o cpu_hog cpu_hog.c
gcc -O2 -Wall -static -o io_pulse io_pulse.c
make -C /lib/modules/6.8.0-47-generic/build M=/home/vishal/OS-Jackfruit/boilerplate modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-47-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
  CC [M]  /home/vishal/OS-Jackfruit/boilerplate/monitor.o
  MODPOST /home/vishal/OS-Jackfruit/boilerplate/Module.symvers
  CC [M]  /home/vishal/OS-Jackfruit/boilerplate/monitor.mod.o
  LD [M]  /home/vishal/OS-Jackfruit/boilerplate/monitor.ko
  BTF [M] /home/vishal/OS-Jackfruit/boilerplate/monitor.ko
Skipping BTF generation for /home/vishal/OS-Jackfruit/boilerplate/monitor.ko due to unavailability of
 vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-47-generic'
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo insmod monitor.ko
ls -l /dev/container_monitor
crw------- 1 root root 510, 0 Apr 23 12:42 /dev/container_monitor
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

1

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boilerplate
sudo ./engine start alpha ~/OS-Jackfruit/rootfs-alpha /bin/sh --soft-mib 48 --hard-mib 80
sudo ./engine start beta ~/OS-Jackfruit/rootfs-beta /bin/sh --soft-mib 64 --hard-mib 96
[sudo] password for vishal:
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ~/OS-Jackfruit/rootfs-base
Supervisor started with base rootfs: /home/vishal/OS-Jackfruit/rootfs-base
Starting container alpha
Started container alpha with PID 53750
Starting container beta
Started container beta with PID 53755
Finished writing logs
Container with PID 53750 exited
Container with PID 53755 exited
Finished writing logs
```

2

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
Container with PID 53750 exited
Container with PID 53755 exited
Finished writing logs
ID      PID     STATE
beta    53755   STOPPED
alpha   53750   STOPPED
```

3

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine logs alpha
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
alpha   53750   STOPPED
Logs for alpha: logs/alpha.log
```

4

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
vishal@vishal:~/OS-Jackfruit/boilerplate$
Stopping container alpha
```

5,6

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo dmesg -w | grep container_monitor
[sudo] password for vishal:
[32624.648369] [container_monitor] Module loaded. Device: /dev/container_monitor
[32663.749004] [container_monitor] Registering container=alpha pid=50702 soft=41943040 hard=67108864
[32664.610951] [container_monitor] PID=50702 RSS=-1 SOFT=41943040 HARD=67108864
[32669.878654] [container_monitor] Registering container=beta pid=50710 soft=41943040 hard=67108864
[32670.754969] [container_monitor] PID=50710 RSS=-1 SOFT=41943040 HARD=67108864
[32784.309370] [container_monitor] Module unloaded.
[34279.672356] [container_monitor] Module loaded. Device: /dev/container_monitor
[34389.132406] [container_monitor] Registering container=alpha pid=53750 soft=50331648 hard=83886080
[34389.139486] [container_monitor] Registering container=beta pid=53755 soft=67108864 hard=100663296
[34389.275511] [container_monitor] PID=53755 RSS=9043968 SOFT=67108864 HARD=100663296
[34389.275557] [container_monitor] PID=53750 RSS=9043968 SOFT=50331648 HARD=83886080
[34390.299499] [container_monitor] PID=53755 RSS=17432576 SOFT=67108864 HARD=100663296
[34390.299524] [container_monitor] PID=53750 RSS=17432576 SOFT=50331648 HARD=83886080
[34391.323628] [container_monitor] PID=53755 RSS=25821184 SOFT=67108864 HARD=100663296
[34391.323632] [container_monitor] PID=53750 RSS=25821184 SOFT=50331648 HARD=83886080
[34392.347498] [container_monitor] PID=53755 RSS=34209792 SOFT=67108864 HARD=100663296
[34392.347522] [container_monitor] PID=53750 RSS=34209792 SOFT=50331648 HARD=83886080
[34393.371493] [container_monitor] PID=53755 RSS=42467328 SOFT=67108864 HARD=100663296
[34393.371506] [container_monitor] PID=53750 RSS=42598400 SOFT=50331648 HARD=83886080
[34394.395514] [container_monitor] PID=53755 RSS=50855936 SOFT=67108864 HARD=100663296
[34394.395572] [container_monitor] PID=53750 RSS=50987008 SOFT=50331648 HARD=83886080
[34394.395586] [container_monitor] SOFT LIMIT container=alpha pid=53750 rss=50987008 limit=50331648
[34395.419547] [container_monitor] PID=53755 RSS=59244544 SOFT=67108864 HARD=100663296
[34395.419570] [container_monitor] PID=53750 RSS=59375616 SOFT=50331648 HARD=83886080
[34396.443501] [container_monitor] PID=53755 RSS=67633152 SOFT=67108864 HARD=100663296
[34396.443512] [container_monitor] SOFT LIMIT container=beta pid=53755 rss=67633152 limit=67108864
[34396.443517] [container_monitor] PID=53750 RSS=67764224 SOFT=50331648 HARD=83886080
[34397.467501] [container_monitor] PID=53755 RSS=76021760 SOFT=67108864 HARD=100663296
[34397.467517] [container_monitor] PID=53750 RSS=76152832 SOFT=50331648 HARD=83886080
[34398.492512] [container_monitor] PID=53755 RSS=84410368 SOFT=67108864 HARD=100663296
[34398.492528] [container_monitor] PID=53750 RSS=84541440 SOFT=50331648 HARD=83886080
[34398.492540] [container_monitor] HARD LIMIT container=alpha pid=53750 rss=84541440 limit=83886080
[34399.516509] [container_monitor] PID=53755 RSS=92798976 SOFT=67108864 HARD=100663296
[34400.539520] [container_monitor] PID=53755 RSS=101187584 SOFT=67108864 HARD=100663296
[34400.539569] [container_monitor] HARD LIMIT container=beta pid=53755 rss=101187584 limit=100663296
[34663.171938] [container_monitor] Registering container=memtest pid=54084 soft=10485760 hard=20971520
[34663.708651] [container_monitor] PID=54084 RSS=9043968 SOFT=10485760 HARD=20971520
[34664.732623] [container_monitor] PID=54084 RSS=17432576 SOFT=10485760 HARD=20971520
[34664.732629] [container_monitor] SOFT LIMIT container=memtest pid=54084 rss=17432576 limit=10485760
[34665.756616] [container_monitor] PID=54084 RSS=25821184 SOFT=10485760 HARD=20971520
[34665.756635] [container_monitor] HARD LIMIT container=memtest pid=54084 rss=25821184 limit=20971520
```

```
Container with PID 54084 exited
ID       PID      STATE
memtest  54084    STOPPED
beta     53755    STOPPED
alpha    53750    STOPPED
```

7

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ # Run CPU-bound workload at normal priority
time sudo ./engine run cpu_normal ~/OS-Jackfruit/rootfs-alpha /cpu_hog

# Run CPU-bound workload at lower priority (nice)
time nice -n 10 sudo ./engine run cpu_low ~/OS-Jackfruit/rootfs-alpha /cpu_hog

real    0m0.033s
user    0m0.014s
sys     0m0.015s

real    0m0.032s
user    0m0.010s
sys     0m0.019s
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine start cpu_hi ~/OS-Jackfruit/rootfs-alpha /cpu_hog
sudo nice -n 19 sudo ./engine start cpu_lo ~/OS-Jackfruit/rootfs-beta /cpu_hog
sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
Running container cpu_normal in foreground
Started container cpu_normal with PID 54470
Finished writing logs
Container cpu_normal finished
Starting container cpu_hi
Started container cpu_hi with PID 54516
Starting container cpu_lo
Started container cpu_lo with PID 54523
ID       PID      STATE
cpu_lo   54523    RUNNING
cpu_hi   54516    RUNNING
cpu_normal        54470    RUNNING
memtest 54084     STOPPED
beta     53755    STOPPED
alpha    53750    STOPPED
Finished writing logs
Container with PID 54523 exited
Container with PID 54516 exited
Finished writing logs
```

8

```
Finished writing logs
Container with PID 54523 exited
Container with PID 54516 exited
Finished writing logs
Stopping container alpha
^C
vishal@vishal:~/OS-Jackfruit/boilerplate$ ps aux | grep engine
sudo dmesg | tail -10
sudo rmmod monitor
vishal      2002  0.0  0.0 172596  7616 ?        Sl   Apr20   0:24 /usr/libexec/ibus-engine-simple
vishal     54634  0.0  0.0  18188  2560 pts/0    S+   12:56   0:00 grep --color=auto engine
[35048.736715] [container_monitor] PID=54523 RSS=67764224 SOFT=41943040 HARD=67108864
[35048.736744] [container_monitor] HARD LIMIT container=cpu_lo pid=54523 rss=67764224 limit=67108864
[35048.736756] [container_monitor] PID=54516 RSS=67764224 SOFT=41943040 HARD=67108864
[35048.736781] [container_monitor] HARD LIMIT container=cpu_hi pid=54516 rss=67764224 limit=67108864
[35049.575393] audit: type=1400 audit(1776929131.186:3769): apparmor="DENIED" operation="open" class="file
" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=36729 comm="MemoryPoller" requested_mask
="r" denied_mask="r" fsuid=1000 ouid=0
[35054.576474] audit: type=1400 audit(1776929136.187:3770): apparmor="DENIED" operation="open" class="file
" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=36729 comm="MemoryPoller" requested_mask
="r" denied_mask="r" fsuid=1000 ouid=0
[35113.414979] audit: type=1400 audit(1776929195.024:3771): apparmor="DENIED" operation="open" class="file
" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=36729 comm="MemoryPoller" requested_mask
="r" denied_mask="r" fsuid=1000 ouid=0
[35118.415560] audit: type=1400 audit(1776929200.025:3772): apparmor="DENIED" operation="open" class="file
" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=36729 comm="MemoryPoller" requested_mask
="r" denied_mask="r" fsuid=1000 ouid=0
[35123.443368] audit: type=1400 audit(1776929205.053:3773): apparmor="DENIED" operation="open" class="file
" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=36729 comm="MemoryPoller" requested_mask
="r" denied_mask="r" fsuid=1000 ouid=0
[35128.448151] audit: type=1400 audit(1776929210.058:3774): apparmor="DENIED" operation="open" class="file
" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=36729 comm="MemoryPoller" requested_mask
="r" denied_mask="r" fsuid=1000 ouid=0
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
sudo ./engine stop beta
sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
vishal@vishal:~/OS-Jackfruit/boilerplate$
```